Frenkel–Kontorova Parameters for an Electron Lattice,

computed from Coulomb rather than fitted

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PROSPECTUS — not a draft. Written August 28, 2026.

Status: this is a plan, not a paper. It records what the claim would be, which two numbers it needs, how to measure them, what controls each must pass, and what is already known. Nothing here is established. It exists so the work can be picked up without reconstructing the reasoning, and so that the claim is not made before the measurements support it.

The claim, if it survives

The Frenkel–Kontorova model takes two parameters as input: the elastic stiffness κ of the chain and the amplitude of the periodic substrate, entering through $K = V''(u_{\min})$. They are normally fitted. RealQM computes both from Coulomb interactions between charge densities, with no adjustable coupling, and with extended deformable domains in place of point masses. Applied to the *electron* subsystem this is a microscopic charge-density-wave model built from an energy functional rather than postulated.

What the two parameters buy is the transport regime, without ever measuring a barrier directly:

$$W = \sqrt{\kappa/K}, \qquad E_{\mathrm{PN}} \sim \exp(-c\, W/a). \tag{1}$$

$W \gg a$ is a wide kink, exponentially small pinning, a sliding electron lattice. $W \sim a$ is a narrow kink and a pinned one. This is the Aubry transition, and it is the question the companion article could not answer by differencing energies — *because* of the exponential in (**??**), which magnifies every error in the two curvatures.

Why this is worth doing at all

The honest competitive position: DFT can also supply CDW parameters from first principles, is calibrated, and computing Peierls barriers that way is routine. The distinctive element here is **not accuracy**. It is that the Frenkel–Kontorova chain is *literal* in this construction — the domains are the particles, one electron each, rather than a caricature of a smooth density. Whether that interpretive advantage is worth a paper is a judgement to make after the numbers exist, not before.

The two measurements

Both are local curvatures taken between configurations that differ infinitesimally, which is the regime where discretisation errors cancel. This is the whole methodological point: the companion article failed because it differenced distant configurations.

K — **the substrate curvature.** Rigid shift of every domain wall by $u = d\,a$ against fixed kernels, at the energy minimum:

$$K = \frac{2\,[E(u_{\min}+\delta) - E(u_{\min})]}{\delta^2}.$$

Requires the minimum to be located first; on the chain it is *not* at $d = 0$ (see below).

κ — **the chain stiffness.** Not yet attempted. Impose a uniform *strain* on the electron spacing with the kernels held fixed — the domains compressed or dilated relative to the lattice — and fit the quadratic. This is a different deformation from the rigid slide: the slide is $u(x) = \text{const}$, the strain is $u(x) \propto x$.

Controls each must pass

1. **Translation.** $E(d{=}0) = E(d{=}1)$ exactly; the two configurations are related by a relabelling. Verified exactly on a ring: 0.56101 at both ends, and two symmetry pairs identical to five decimals.
2. **Reflection.** $E(d) = E(-d)$ through a core, hence symmetry about $d = \frac{1}{2}$. The seven-core chain *fails* this — its minimum sits beyond $d = 0.5$ — because two end domains are 29% of the system. This is the outstanding obstacle.
3. **Constancy of the curvature** across the smallest few displacements. A drift in K with δ means the quadratic regime has not been reached.
4. **Reproducibility** to five decimals on repeated runs, which the frozen-partition method delivers when the diagnostic poller is off.

What is already known

quantity	value	status
K at the commensurate registry $d = 0$	-0.089 Ha/a_0^2	four displacements agree to 10%
K at the energy minimum	$\approx +0.27$ Ha/a_0^2	contaminated by chain ends
implied polarisability $1/K$	≈ 3.8 a_0^3/electron	hydrogen-like, plausible
equilibrium at $r_c = 1.93$	$a = 5.43$ a_0	five-point scan, clean
κ	—	not attempted

The negative K at $d = 0$ is the more solid of the two and is itself a result: the commensurate registry, every domain centred on its own kernel, is *unstable* to a uniform shift. The stable configuration is bond-centred, which agrees with the registry preference found independently in the companion article.

The next run, concretely

Rigid slide at $n = 11$, $r_c = 1.93$, $a = 5.5$, $h \approx 0.35$ (a 186^3 grid), sampling $d = 0.375, 0.4375, 0.5, 0.5625, 0.625$. The question is whether the minimum returns to $d = \frac{1}{2}$ once the end domains fall from 29% to 18% of the system. If it does, the end effect is confirmed, K follows there, and $n = 15$ gives the extrapolation. If it does not, the asymmetry is something else and the programme stops until that is understood.

Known limitations, to be stated in any paper

- **Staircase floor.** The partition assigns whole cells, so displacements below one cell flip no labels: K cannot be evaluated nearer the minimum than $u \approx 0.3\ \mathrm{a}_0$ without sub-cell interface resolution. Two configurations 0.125 and 0.250 of a spacing apart returned *identical* energies to five decimals, which is this effect caught directly.
- **Long-range coupling.** Equation (**??**) assumes nearest-neighbour springs. Coulomb is long-ranged, giving power-law rather than exponential kink tails and shifting the transition. A guide to the regime, not a formula good to three figures.
- **No zero-point motion.** Static RealQM has no quantum fluctuation of the electron chain, so it returns the *classical* FK parameters and necessarily over-pins relative to the real system. This is the same over-localisation that exact diagonalisation exposes on the same chains, where the hole occupies four or five sites and this construction admits one.